KSRTC

WORKPLACE PERFORMANCE MEASUREMENT

Deep-Learning Activity Detection

ResNet-18 CNN trained on 2,000 labelled video frames



Hold-out test accuracy • 93.0%

Waybill Entry Desk • KSRTC Depot • Phase-I Evidence Module

Method after Nesterak, Szelagowski & Radziszewski, J. Intell. Manuf. (2025)

Methodology — from video to automatic detection

Phase I / II evidence chain built for the KSRTC waybill-entry work station

1

4K IP-Camera Capture
65 min 51 s film of the waybill-entry desk
23.04 GB, 1920 x 1080 @ 25 fps

RECORD

2

Frame Extraction
1 frame per second
3,951 frames from the raw footage

SAMPLE

3

Human Annotation
2,000 frames labelled by 4 activities
write / crew / check / wait, custom label tool

LABEL

4

ResNet-18 Transfer Learning
ImageNet weights, 4-class head, class-balanced
train 75% / val 15% / test 10%, Adam lr 3e-4

TRAIN

5

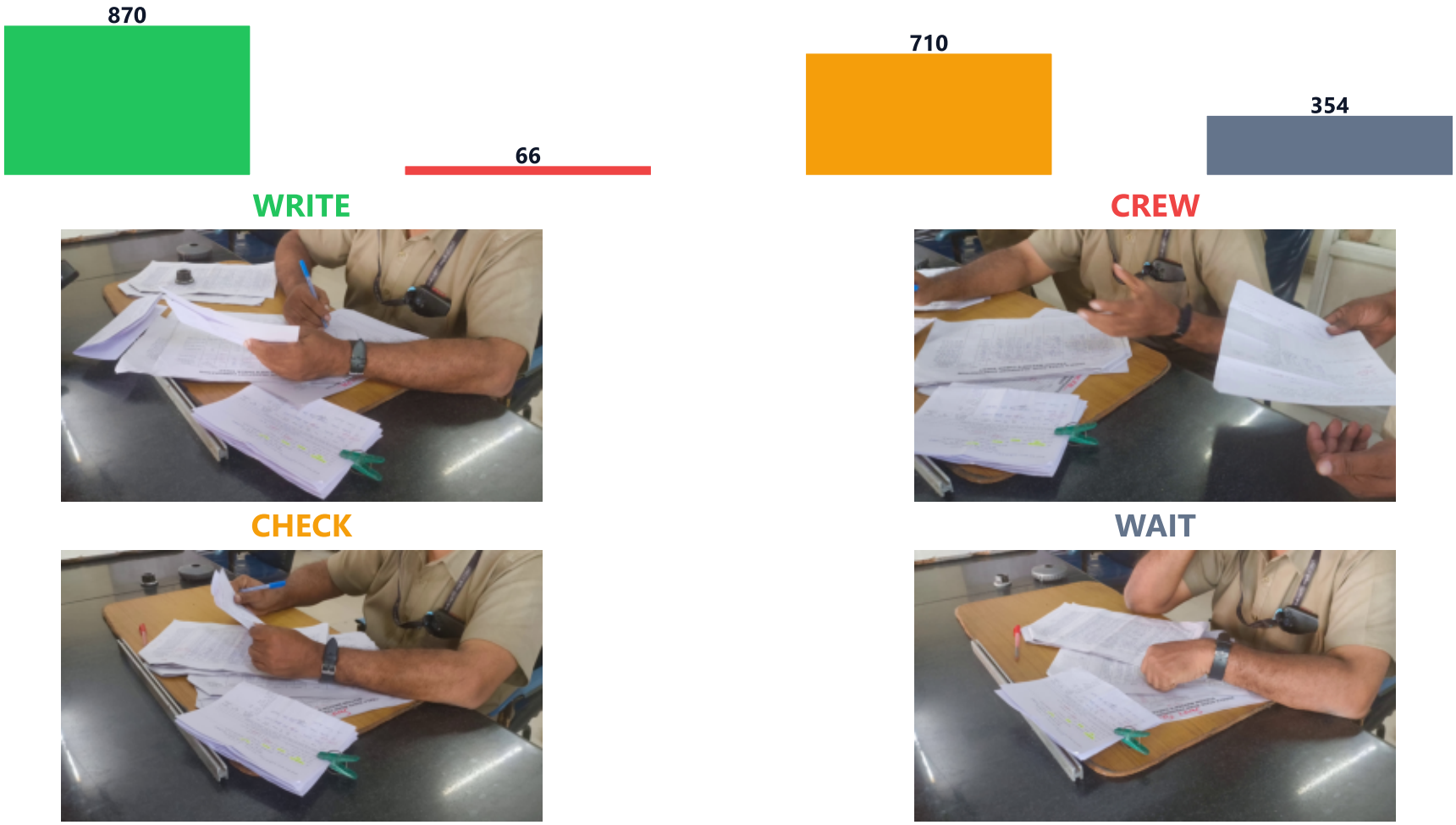
Hold-out Test
93.0% accuracy, per-class precision/recall
+ confusion matrix analysed

EVALUATE

6

Web-App Upload + Detect
Upload frames to the Flask dashboard,
CNN auto-labels them into an event log

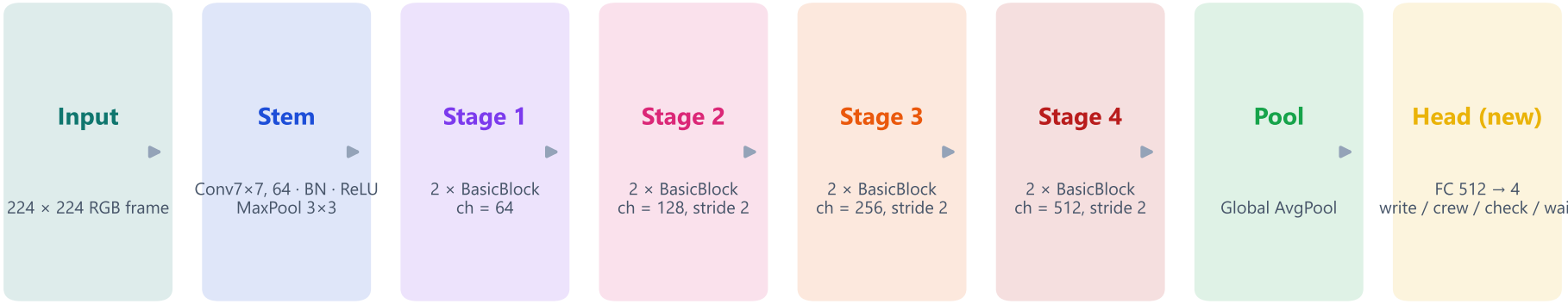
DEPLOY



Labelled Training Data

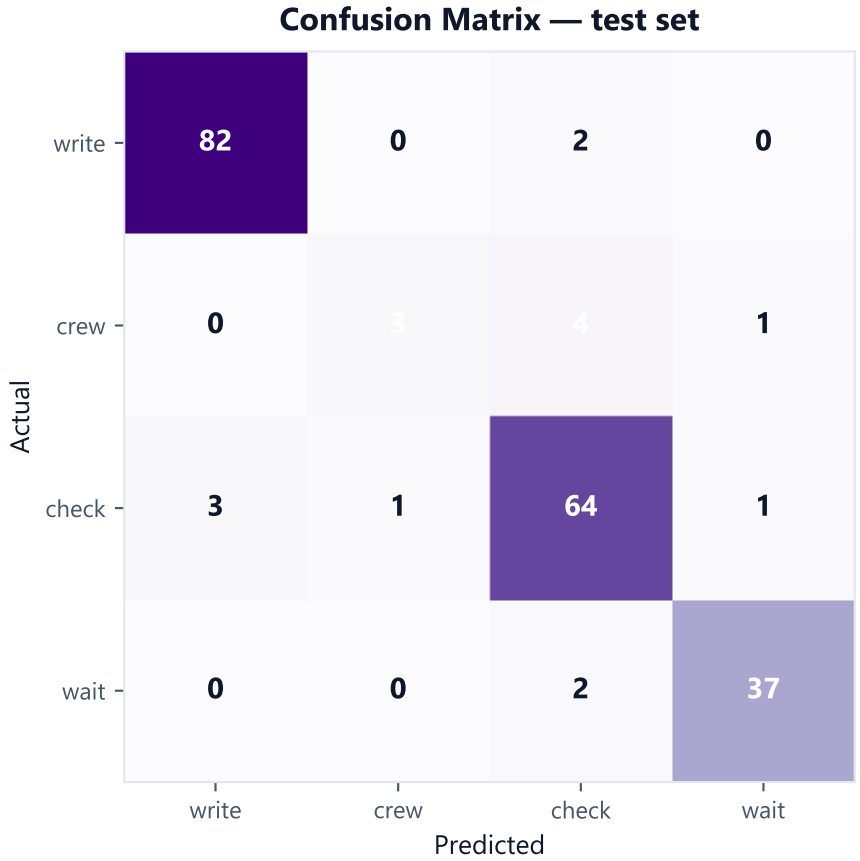
2,000 frames from D:\ksrtc\frames clear classified into four activities at 1 s resolution.

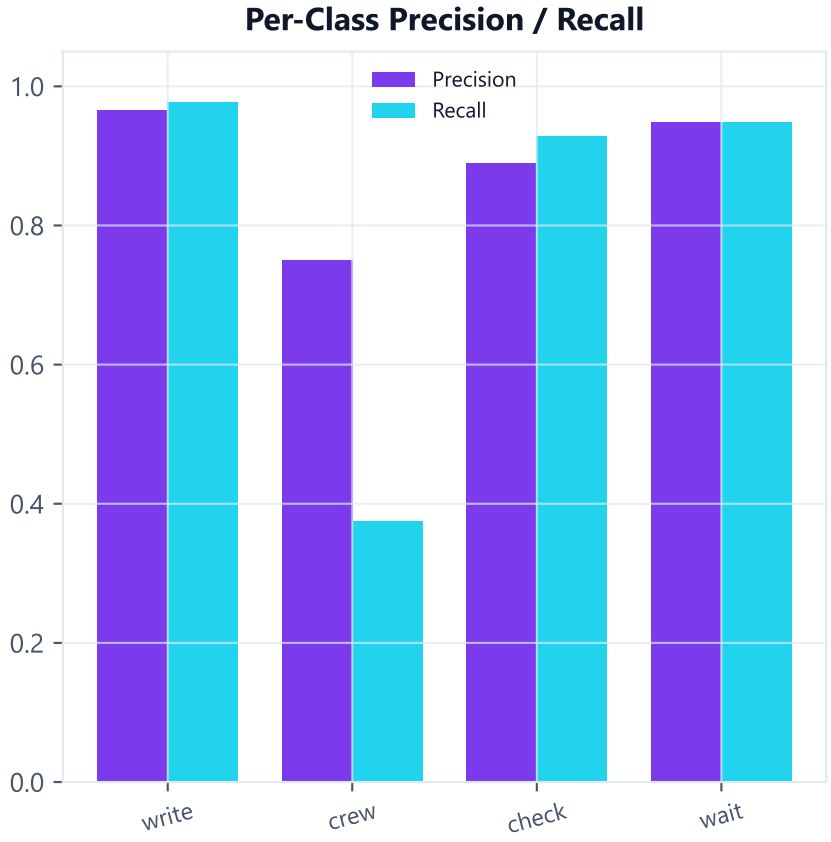
ResNet-18 Architecture for Frame Classification



AUGMENTATION	RandomHorizontalFlip (p=0.3) · ColorJitter · Resize 224
SAMPLING	WeightedRandomSampler — over-samples the rare crew class
LOSS / OPT	CrossEntropyLoss (class-weighted) · Adam 3e-4 · ReduceLROnPlateau
SCHEDULE	25 epochs, batch 24 — best model kept by validation accuracy

Transfer learning: ImageNet-pretrained trunk, head replaced for 4 WPM activities



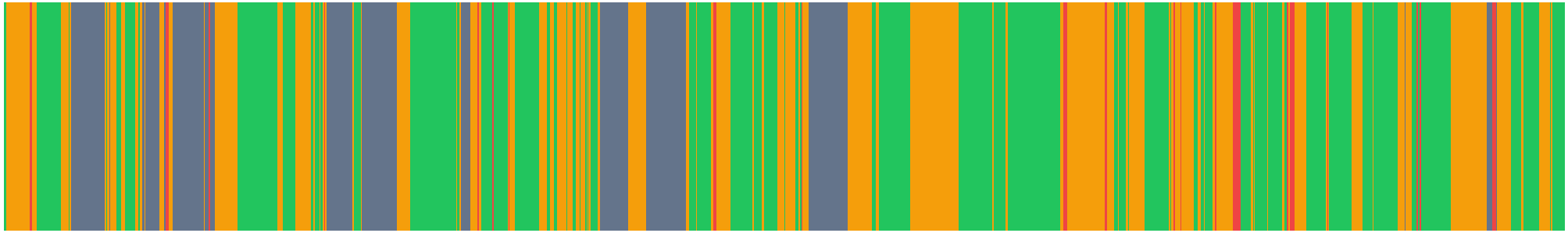


93.0% **Hold-out Test Accuracy**

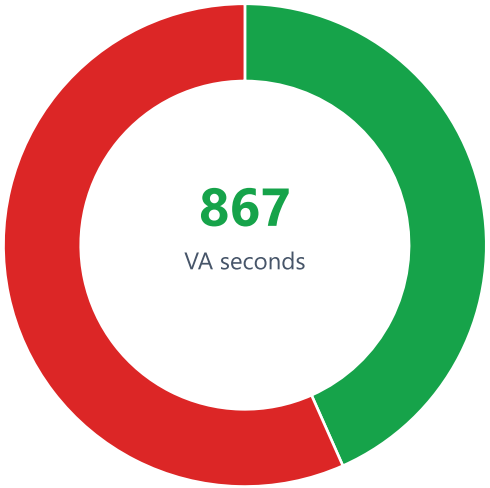
200 test frames · 2000 labelled frames · crew correctly caught despite only 66 samples

Model: cnn_resnet18.pt · resnet18

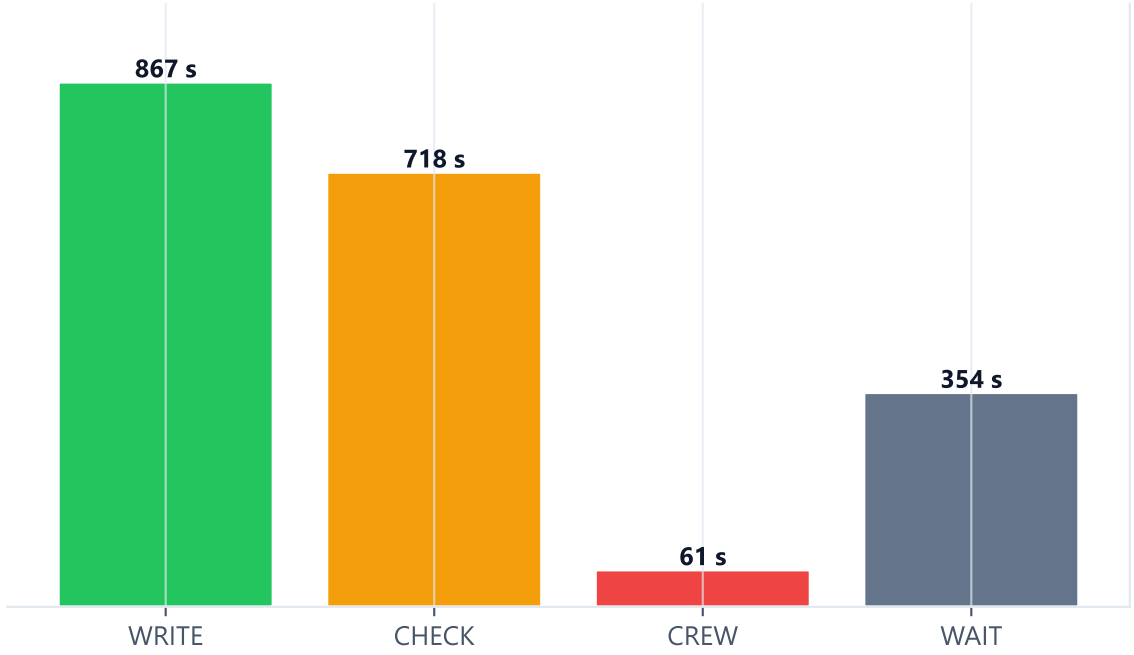
Predicted Activity Timeline — 2,000 s of observation



VA / NVA split



Activity seconds — automatic detection over the full observation



A total of 2000 frames processed, 187 duty events segmented, VA (value-added) = 43.4 %, NVA = 56.6 % — identical logic to the automation module used by the dashboard.

Web-App Integration — /ml/cnn upload & detect

Upload frame images (or a .zip) on the live dashboard; the CNN labels them instantly

Flask route

/ml/cnn + POST /ml/cnn/predict

cnn.html template

Modules/cnn_module.py · CnnDetector

Upload

multiple .jpg/.png or .zip
MAX_CONTENT_LENGTH 256 MB

saved under static/cnn_uploads

zip auto-extracted

Inference

ResNet-18 forward pass
224 × 224, batch 32

per-frame class + VA/NVA type

returned as JSON

Results

class distribution
event timeline table

per-frame prediction list

VA / NVA split summary

Live dashboard page offers:

- KPI strip — model ONLINE, test accuracy, classes, frames uploaded
- Confusion matrix & per-class precision/recall rendered in the dashboard theme
- Upload + run button, then an event table, class distribution and VA/NVA result
- Everything auto-logs into the same duty-log pipeline used by the dashboard

Trained locally with scripts/train_cnn_resnet.py, shipped as data/cnn_resnet18.pt